

H-LAYER analytic closure: a consolidation of the T-016 to T-021 arc, and

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

B1-RH-ENUM (the Reading-H enumerated selection) and B2-PROPA-HLAYER (Proposition A) rest, after the SC-SCOPE lift (2026-06-09), on the SOLE hypothesis H-LAYER: the isotropic Gaussian–Hartree variational layer is the comparison infimum. H-LAYER’s open content was STEP-5B – the beyond-diagonal, beyond-second-cumulant refinements for NON-enumerated patterns. The T-016 to T-021 arc closes those refinements class-wide. This note is a chronicle + status consolidation (in the style of the SC-SCOPE and DR-2 programme consolidations); it introduces no new result and formalises the one remaining operator item. Scope: operating endpoint $I = 2 \times 10^{-3}$, anchor $\mu^2 = 5 \times 10^{-3}$, ROBUSTNESS-MU2 band $[\times 0.5, \times 2]$.

2. The H-LAYER decomposition

The isotropic layer is the infimum iff no admissible competitor lowers the free energy. The comparison splits into:

- (D) **Diagonal sector** – is the isotropic dressing the diagonal-class minimum? (Math427; H-diag.)
- (O) **Off-diagonal sector** – can an off-diagonal (Bloch) covariance lower F ? (RES-1; the condensate-induced Bogoliubov bands.)
- (S) **Selection floor** – does the STEP-5B layer-margin survive the third-cumulant endpoint correction for all competitors? (SC-SCOPE; the additive- energy floor $K_{\text{floor}} \leq T'$.)

3. The arc (T-016 to T-021)

T-016 (isotropy-infimum-core v1.1) *Diagonal (D)*. The condensate-free Hessian is BLOCK-DIAGONAL in angular momentum: the rotation-invariant interaction curves only the $l = 0$ breathing sector ($\Phi''_{\text{diag}} = \frac{3}{2}u_{\text{eff}} = +4.03 > 0$); every $l \geq 1$ sector sees the entropy Hessian $\frac{1}{2}G_*^{-2} \geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$ (ANALYTIC infimum at the gap shell). The isotropic dressing is a STRICT, curvature-protected diagonal minimum.

T-017 (res5-chi-link-bypass v1.1) *Selection floor (S)*. The weighted Lemma A (Cauchy–Schwarz, $r(t) \leq T'$ for $t \neq 0$) gives $K_{\text{floor}}(c) = \sum_{t \neq 0} w_t^2 / I^2 \leq T'(Q)$ for ANY amplitudes – the $\chi(P)$ carrier-richness link is BYPASSED. With the admissible pin $\max_Q T' = 13$, $K_{\text{floor}} \leq 13 < 20.5 < 26.2$ class-wide.

- T-018** (hdiag-offdiag-gershgorin-threshold v1.1) *Off-diagonal (O)*. The condensate block B^{cond} is DIAGONAL in Q (independent channels), so $\|E^{-1/2}B^{\text{cond}}E^{-1/2}\| = \text{max-channel ratio} = R_{\text{lead}}(Q) \leq 0.650$ class-wide ($R_{\text{lead}} = 23.2(1 + K_{\text{floor}})I$, $K_{\text{floor}} \leq T' \leq 13$).
- T-019** (offdiag-exchange-scalar-identification v1.0) *Off-diagonal (O)*. For TECT's LOCAL interaction there is NO A-independent attractive Fock exchange: the mean-field off-diagonal Hessian is the stabilising density-density term $+\frac{3}{2}u_{\text{eff}} > 0$ (Hartree=Fock for a local functional). The genuine off-diagonal obstruction is the A-dependent bubble (T-018) + the beyond-mean-field two-loop = SC-SCOPE.
- T-020** (classwide-secondcumulant-stability v1.0) *Off-diagonal (O)*. The Bogoliubov band is positive iff $\rho R_{\text{lead}} < 1$; at $\rho = 1$ (no-credit) this is $R_{\text{lead}} < 1$, class-wide ≤ 0.650 (T-018). The screening $\rho \leq 1$ improves it. Second-cumulant off-diagonal stability is CLASS-WIDE.
- T-021** (scscope-classwide-endpoint v1.0) *Selection floor (S)*. The certified joint reduces (MARGIN-cancellation) to joint(ρ_{lat}), monotone, with joint $> 1 \iff \rho_{\text{lat}} > 4.347 \iff T' < 60.4$. The pin $T' \leq 13 < 60.4$ closes the SC-SCOPE third-cumulant endpoint class-wide (joint $\geq \times 1.095$ over the μ^2 -band).

4. Final analytic status

(D) diagonal isotropy	STRICT minimum, analytic (T-016)
(O) off-diag, A-independent	stabilising $+\frac{3}{2}u_{\text{eff}}$ (T-019)
(O) off-diag, A-dependent, 2nd cumulant	class-wide stable $\rho R_{\text{lead}} \leq 0.650$ (T-018,020)
(S) floor $K_{\text{floor}} \leq T' \leq 13$	any amplitudes (T-017)
(S) SC-SCOPE 3rd cumulant	class-wide joint $\geq \times 1.095$ ($T' < 60.4$) (T-021)
remaining	operator competitor-class definition

Every analytic H-LAYER residual identified in the arc is closed CLASS-WIDE within the admissible crystallographic class, at the operating endpoint, over $[\times 0.5, \times 2]$.

5. The remaining item, formalised (for operator sign-off)

The closures quantify over the *admissible competitor class* $\mathcal{A}_{\text{adm}}$. Formalised:

$$\mathcal{A}_{\text{adm}} = \left\{ Q \subset \Lambda \cap \{|x|^2 = R\} : |Q| \leq n_{\text{pack}}, \text{ pairwise } \angle(k_i, k_j) \geq \theta_{\min}(\hat{r}) \right\},$$

i.e. finite subsets of crystallographic momentum-shell lattice points (rational shell $\Lambda \cap \{|x|^2 = R\}$), bounded by the packing $n_{\text{pack}} = 40.7$ and the coherence-resolution separation $\theta_{\min}$ (H-ADM-COH, discharged@lattice via R-026/R-027/R-028). On $\mathcal{A}_{\text{adm}}$ the sum-circle richness satisfies $T'(Q) \leq 13$ (T-017 pin) and the arc closes. **The operator item** is the COMPLETENESS of $\mathcal{A}_{\text{adm}}$: that every physically relevant B1 Reading-H competitor lies in $\mathcal{A}_{\text{adm}}$ (crystallographic, separated, $\leq n_{\text{pack}}$). This is a MODELLING statement, not an analytic estimate; its sign-off (or a refined definition) is the operator decision. Its falsification: a physical competitor OUTSIDE $\mathcal{A}_{\text{adm}}$ (non-lattice, sub- $\theta_{\min}$, or super-packing) that lowers F below the isotropic layer.

6. Path to a re-tier (operator decision)

If the operator signs off on $\mathcal{A}_{\text{adm}}$ completeness, the H-LAYER hypothesis becomes a PROVED-class statement on $\mathcal{A}_{\text{adm}}$ (analytic residuals discharged), and B1/B2 may be re-examined for a tier move from T6 CONDITIONAL on {H-LAYER}. The re-tier still inherits: (i) the thin-certified $R_{\text{max}} = 0.385$ (Parseval-pinned quartic) and sunset $S = 1.13$; (ii) the T-017 $T' \leq 13$ pin; (iii) the anchoring $u_{\text{eff}} > 0$, $g \geq 0$ on $[\times 0.5, \times 2]$. These are STRONG-EVIDENCE / thin-certified, not unconditional theorems, so the honest ceiling for B1/B2 after sign-off is a NAMED-HYPOTHESIS-REDUCED T6 (H-LAYER $\rightarrow \mathcal{A}_{\text{adm}}$ -completeness + the inherited grades), NOT an unconditional T7. No tier flip is enacted here.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*This claims H-LAYER is closed.*” CALIBRATED: it claims the ANALYTIC residuals are class-wide closed; the competitor-class completeness (a modelling statement) and the inherited thin grades remain. The status box (§4) separates the closed analytic rows from the remaining operator row.
- (β) “*The closures are anchored / thin.*” ACKNOWLEDGED: every row inherits the operating-endpoint scope and the ROBUSTNESS-MU2 band; SC-SCOPE inherits the thin R_{max} certificate. The consolidation does not upgrade any grade; it records them.
- (γ) “*Does this flip B1/B2?*” DISMISSED: no. §6 states the re-tier is the operator decision and is ceilinged at a hypothesis-reduced T6, not T7. B1/B2 stay T6 on {H-LAYER}.

Quantitative sanity check (mandatory): (D) $\Phi''_{\text{diag}} = 4.03 > 0$, entropy $\geq 4.64 \times 10^{-2}$; (O) $R_{\text{lead}} \leq 0.650 < 1$, $+3/2 u_{\text{eff}} > 0$; (S) $T' \leq 13 < 60.4$, joint $\geq \times 1.095$; *scope* – operating endpoint, $[\times 0.5, \times 2]$; all figures cited to T-016..T-021.

8. Result footer

Result ID:	B2-PROPA-HLAYER / H-LAYER analytic-closure consolidation
Precise statement:	The T-016..T-021 arc closes every ANALYTIC H-LAYER residual class-wide: (D) diagonal isotropy strict minimum (T-016); (O) no A-independent Fock exchange, $+3/2 u_{\text{eff}}$ stabilising (T-019), A-dependent 2nd-cumulant stability $\rho \cdot R_{\text{lead}} \leq 0.650$ class-wide (T-018,020); (S) $K_{\text{floor}} \leq T' \leq 13$ any amplitudes (T-017), SC-SCOPE 3rd-cumulant endpoint joint $\geq \times 1.095$ ($T' < 60.4$) class-wide (T-021). Remaining = operator competitor-class definition: A_{adm} = crystallographic-shell subsets, separated, $\leq n_{\text{pack}}$, on which $T' \leq 13$. Completeness of A_{adm} is a modelling sign-off, not an analytic estimate.
Scope:	Operating endpoint $I = 2e-3$; ROBUSTNESS-MU2 $[\times 0.5, \times 2]$. Consolidation (no new claim). Inherits thin-certified $R_{\text{max}} = 0.385$ + sunset $S = 1.13$ + T-017 pin.
Dependencies:	isotropy-infimum-core v1.1 (T-016); res5-chi-link-bypass v1.1 (T-017); hdiag-offdiag-gershgorin-threshold v1.1 (T-018); offdiag-exchange-scalar-identification v1.0 (T-019); classwide-secondcumulant-stability v1.0 (T-020); scscope-classwide-endpoint v1.0 (T-021).
Evidence grade:	CONSOLIDATION (cites executed results T-016..T-021, each 3-6/6).
Reproduction command:	(per-row) see the cited note's script; this note adds no compute.

Falsification gate: A physical B1 competitor OUTSIDE A_adm (non-lattice, sub-theta_min, or super-n_pack) lowering F below the isotropic layer; or a re-opened analytic row (any of D/O/S above).

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. Records that the analytic H-LAYER residuals are class-wide closed; the re-tier is the operator competitor-class decision, ceilinged at a hypothesis-reduced T6 (NOT T7).

No-overclaim statement: Does NOT flip B1/B2 and does NOT claim unconditional closure. The analytic residuals are class-wide closed; the competitor-class completeness (modelling) and the inherited thin-certified grades (R_max, sunset, T-017 pin, anchoring) remain. B1/B2 stay T6.

Next required action: OPERATOR: sign off on (or refine) the admissible competitor class A_adm completeness. On sign-off, B1/B2 may be re-examined for a hypothesis-reduced T6 (H-LAYER -> A_adm-completeness + inherited grades).